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SUPERSEDING
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FEDERAL SPECIFICATION

SOLDERING IRON, SELF-HEATING (WITHOUT ELECTRIC OR EXTERNAL HEAT)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers self-heating soldering irons without electric or external heat applied to the soldering tip. The heat is furnished by the firing of a single-shot cartridge unit loaded with a chemical thermal charge. This tool is useful where electric power is not available or convenient, or where use of a blowtorch would be hazardous.

1.1.1 *Federal specification coverage.*—Federal specifications do not include all types and grades of the commodities indicated by the titles of the specifications, but include only those most commonly used by the Federal Government.

1.2 Classification.

1.2.1 *Type and grades.*—The soldering irons covered by this specification shall be of but one type furnished in the following grades, as specified (see 6.1):

Grade A.—Corrosion-resisting steel body assembly with plastic handle.

Grade B.—Metal body assembly with wood handle.

1.2.2 *Soldering tips.*—Soldering tips shall be furnished in styles and sizes specified in 3.4 or in the invitation to bid, contract, or order (see figure 2, table I and 6.1).

1.2.3 *Heating units.*—Heating units shall be of one type in accordance with 3.7 and furnished in the quantities specified in the invitation to bid, contract, or order (see 6.1).

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standard, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

QQ–S–763—Steel Bars, Shapes and Forgings—Corrosion-Resisting.

QQ–S–766—Steel, Corrosion-Resisting; Plates, Sheets and Strips.

Military Specifications:

MIL–P–14—Plastic-Materials, Molding, and Plastic-Parts, Molded; Thermosetting.

MIL–P–15424—Packaging of Hand Tools; General Specification for Domestic and Overseas Shipment and Storage.

Military Standard:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Illustrations.—The illustrations included herein are for the purpose of identification and are not intended to preclude the purchase of soldering irons which are otherwise in accordance with the requirements of this specification.

3.2 Design.—The self-heating soldering iron covered by this specification shall be of the latest design, convenient to operate and entirely adequate for soldering wire connections, joining seams for large and small surface areas and similar applications It shall be a self-contained heating unit and operate without electric current or blowtorch.

3.2.1 *Interchangeability.*—All replaceable parts and components shall be constructed to definite standards, tolerances and clearances in order that they may be replaced or adjusted without requiring modification.

3.3 Material.—Material shall be as herein specified. Material not definitely specified shall be of good commercial quality and entirely suitable for the purpose intended. Material shall be free from defects and imperfections that might adversely affect the serviceability of the finished product.

3.4 Soldering iron unit.—Unless otherwise specified (see 6.1), the iron shall be furnished complete with one wire brush and one each of the ⅜-inch pyramid and chisel-type tips.

3.4.1 *Wire brush.*—The brush shall be of appropriate diameter for removing residue from the cartridge chamber of the body assembly after firing several cartridges.

3.5 Components.—The self-heating soldering iron covered by this specification shall consist of a body assembly, handle, reflector, firing pin mechanism, tips and heating units (see figure 1).

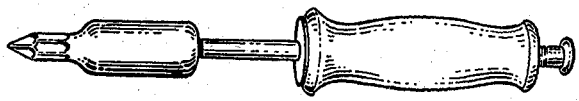


FIGURE 1.—Soldering iron.

3.6 Soldering tip.—The tips shall be made of a pure electrolytic, oxygen-free copper with working surfaces suitably formed. The tips shall be designed and constructed to permit interchangeable attachment of the various styles and sizes with the body assembly. The design shall be either of the pyramid tip style, for soldering wire and similar work; or the chisel tip style, for soldering flat and large surfaces (see figure 2). Tips shall conform to table I for the size and style specified (see 3.4 and 6.1).

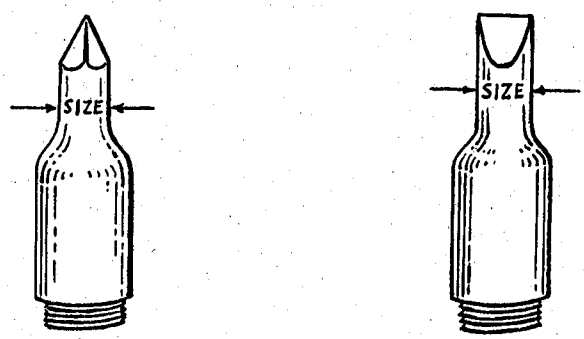


FIGURE 2.—Soldering iron tips.

TABLE I.—*Soldering tips, styles, and sizes*

Style	Size, diameter
	Inch
Pyramid	⅜, ⅝
Chisel	⅜, ⅝, 1

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3.7 Heating unit.—The heating unit shall be a single shot, self-contained cartridge with a chemical thermal charge and percussion cap for priming, or any other equally effective self-contained heating unit. It shall be designed for insertion in the body assembly in a manner to make proper contact with the firing pin. It shall supply the energy required for heating the soldering tip to a minimum temperature of 356°F. (180°C.) in 30 seconds or less, and maintain a soldering temperature for at least 6 minutes at an ambient temperature of 75°F., when tested as specified in 4.3. At minus 40°F., the unit with a ⅜-inch pyramidal-shaped tip, shall reach initial solder melting temperature of 356°F. in 30 seconds or less and maintain a soldering temperature for at least 3 minutes when tested as specified in 4.4. If the heating unit is of the chemical percussion priming type the cartridge shall be nonexplosive without a percussion contact primer and have an ignition point of 600°F. The primer shall be inset from the plane of the cartridge head as a precaution against accidental discharge. Cartridges are packaged 12 to a box (see 6.1).

3.8 Identification marking.—Each iron shall be marked in a permanent and legible manner with the manufacturer's name or with a trade-mark of such known character that the source of manufacture may be readily determined.

3.9 Grade A, corrosion-resisting steel body assembly with plastic handle.

3.9.1 *Body.*—The body assembly shall be of corrosion-resisting steel conforming to the requirements of Federal Specification QQ–S–763 (class 7, type A), and shall be of sufficient strength and rigidity to support the forces exerted in applying the iron in the soldering operation. The body shall support and guide the firing mechanism. Provisions shall be made for attaching the different styles of tips and for securing the handle to the body.

3.9.2 *Handle.*—The handle shall be of black thermosetting plastic conforming to the requirements of Military Specification MIL–P–14 (type CFG), and shall be securely fastened to the body assembly. The exterior of the handle shall be smooth and comfortable to the hand of the operator.

3.9.3 *Reflector.*—A reflecting disc approximately 1.125 inches in outside diameter and 0.0625 inch thick shall be located at the tip end of the handle, and shall be corrosion-resisting steel conforming to the requirements of Federal Specification QQ–S–766 (class 2, condition a, finish 7).

3.9.4 *Firing pin mechanism.*—The firing pin mechanism shall be a self-contained mechanism guided by the body assembly, designed and constructed to be manually discharged for release of the thermal charge to heat the iron. The firing pin mechanism shall be of corrosion-resistant steel conforming to Federal Specification QQ–S–763 (class 7, type A).

3.10 Grade B, metal body assembly with wood handle.

3.10.1 *Body assembly.*—The body assembly shall conform to 3.9.1 except any type of metal that will meet the strength and rigidity requirements may be used.

3.10.2 *Handle.*—The handle shall be of wood, securely fastened to the body assembly, shall provide a comfortable handgrip and shall have a protective coating of paint or enamel.

3.10.3 *Reflector.*—The reflector shall conform to 3.9.3 except it shall be of any type of bright-finish metal.

3.10.4 *Firing pin mechanism.*—The firing pin mechanism shall be a self-contained mechanism guided by the body assembly, designed and constructed to be manually discharged for release of the thermal charge to heat the iron.

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3.11 Workmanship.—The iron shall be new and of the quality workmanship necessary to produce a tool of good appearance and of efficient operating and productive characteristics found in first class commercial quality items of a similar nature.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling for lot acceptance.

4.1.1 *Inspection lot.*

4.1.1.1 *Soldering irons.*—All soldering irons of the same grade presented at one time shall be considered a lot for purposes of inspection and tests.

4.1.1.2 *Cartridges.*—All cartridges presented at one time shall be considered a lot for purposes of inspection and tests.

4.1.2 *Sampling for visual and dimensional inspection.*—A random sample of irons or cartridges shall be taken from each inspection lot of material offered for Government inspection of visual and dimensional characteristics with lot acceptance based on the following sampling inspection requirements in accordance with Military Standard MIL–STD–105.

TABLE II.—*Sampling for visual and dimensional inspection A.Q.L. (approx.) = 1.5 percent defective*

Number of irons or cartridges in inspection lot	Number of irons or cartridges in sample	Acceptance number, defective	Rejection number, defective
15 and under.	7	0	1
16 to 40	10	0	1
41 to 110	15	0	1
111 to 300	25	1	2
301 to 500	35	1	2
501 to 800	50	2	3
801 to 1300 ...	75	3	4
1301 and over .	110	4	5

NOTE.—The Government inspector may institute tightened or reduced inspection in accordance with section 9 of Military Standard MIL–STD–105.

4.1.3 *Sampling for tests.*—A random sample of irons or cartridges shall be taken from each inspection lot in accordance with table III and shall be subjected to the test specified in 4.4.

TABLE III.—*Sampling for test*

Number of irons or cartridges in lot	Number of irons or cartridges in sample	Acceptance number, defective	Rejection number, defective
15 and under ..	3	0	1
16 to 40	5	0	1
41 to 110	7	0	1
111 to 300	10	0	1
301 to 500	15	1	2
501 to 800	25	2	3
801 and over ..	35	3	4

4.2 Visual and dimensional inspection.—Each of the sample irons or cartridges shall be visually and dimensionally inspected by the Government inspector to verify compliance with the requirements of this specification. Any item in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective items in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3 Lot acceptance tests.—Each of the sample irons and cartridges taken in accordance with table III shall be tested in accordance with 4.4 to verify compliance with the requirements of this specification. The sample cartridges shall be divided among the sample irons for testing when they are tested at the same time.

4.4 Test procedure.—A test shall be conducted with each style and size of tip being procured by using ⅛-inch diameter ASTM 50 general purpose solder to determine the melting qualities of the iron as specified in 3.7. Failure to meet this requirement shall be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Nonmilitary requirements.

5.1.1 *Preservation.*—Cleaning and preservation shall be in accordance with the manufacturer's standard practice.

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5.1.2 *Packaging.*—Commercial packages will be acceptable.

5.1.3 *Packing.*—The equipment shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose in such manner as to insure acceptance and safe delivery to the designated point by common or other carrier at the lowest rate of the carrier.

5.1.4 *Marking.*

5.1.4.1 *Interior containers.*—Packages and interior containers shall be marked with the name of the material and the name or trademark of the contractor of such known character that the source of supply may be readily determined.

5.1.4.2 *Exterior containers.*—Shipping containers shall be marked with the name of the item, the grade, and the quantity therein, as defined by the contract or order under which shipment is made, the name of the contractor, and the number of the contract or order.

5.2 Military requirements.

5.2.1 *Preparation for shipment.*—Cleaning, preservation, packaging, packing, and marking for shipment shall be in accordance with Military Specification MIL–P–15424.

6. NOTES

6.1 Ordering data.—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Grade of soldering iron required (see 1.2.1).
(c) Style and size of tips required if other than specified in 3.4 (see 1.2.2, 3.6, and table I).
(d) Quantity of heating units required (see 1.2.3 and 3.7).
(e) Whether preparation for domestic or overseas shipment is required (see section 5).

6.1.1 *Data with bid.*—Each bidder shall be required to furnish with his bid complete descriptive data covering the items he proposes to furnish.

6.2 Illustrations.—The illustrations herein are for the convenience of identification and are not intended to preclude the purchase of soldering irons which are otherwise in accordance with the requirements of this specification.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:

Army—Sig E O C
Navy—Sh A

Copies of this specification may be purchased for 10 cents each.

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